Phase 1 - Dynamic Persona Scoring Output System

1. Core Data Structure (UIN-Centric)

Primary Output Table: `doctor_persona_scores`

UIN	full_name	specialty	dr_city	dr_state	academic_score	incentive_score	patient_score	te
D001	Dr. John Smith	Cardiology	Mumbai	Maharashtra	85	23	67	45
D002	Dr. Priya Sharma	Pediatrics	Delhi	Delhi	34	78	92	56

Key Calculated Fields:

- **Primary Persona**: Highest scoring persona bucket
- **Secondary Persona**: Second highest (only if >30 points and within 20 points of primary)
- **Confidence Level**:
  - High: Primary score >70 and >30 points gap from secondary
  - Medium: Primary score 50-70 or 15-30 points gap
  - Low: Primary score <50 or <15 points gap

2. Basic Filter System

Filter Categories:

```
Geographic Filters:
├── State (dropdown: Maharashtra, Delhi, Karnataka, etc.)
├── City (dependent dropdown based on state)
└── Region (Metro, Tier-1, Tier-2, Rural)

Professional Filters:
├── Specialty (Cardiology, Pediatrics, Orthopedics, etc.)
├── Years of Experience (0-5, 5-10, 10-15, 15+ years)
├── Institution Type (Government, Private, Academic)
└── Qualification Level (MBBS, MD, DM, PhD, etc.)

Persona Filters:
├── Primary Persona (dropdown: Academic, Patient-focused, etc.)
├── Secondary Persona (dropdown + "None" option)
├── Confidence Level (High, Medium, Low)
└── Score Range Sliders (0-100 for each persona bucket)

Data Quality Filters:
```

```
├── Profile Completeness (>90%, 70-90%, 50-70%, <50%)
├── Last Updated (Last week, month, quarter, year)
└── Data Sources (Number of sources: 1-3, 4-7, 8+)
```

Sample Filter Interface:

```
[State: All ▼] [City: All ▼] [Specialty: Cardiology ▼]
[Primary Persona: Academic ▼] [Experience: 10-15 years ▼]
[Academic Score: 70-100] [Patient Score: Any]
[Show only High Confidence: ☑]

Results: 234 doctors found
```

3. Trait Analysis (PnC - Persona Combinations)

PnC Scoring System Recommendation:

Method 1: Weighted Combination Score

```python
def calculate_trait_score(persona_scores, combination):
    """
    Example: Academic + Patient focused combination
    combination = ['academic', 'patient']
    """
    if len(combination) == 2:
        # Dual persona combination
        primary_weight = 0.7
        secondary_weight = 0.3

        scores = [persona_scores[combo] for combo in combination]
        scores.sort(reverse=True)  # Highest first

        trait_score = (scores[0] * primary_weight + scores[1] * secondary_weight)
        return min(trait_score, 100)

    elif len(combination) == 3:
        # Triple persona combination
        weights = [0.5, 0.3, 0.2]
        scores = sorted([persona_scores[combo] for combo in combination], reverse=True)
        trait_score = sum(score * weight for score, weight in zip(scores, weights))
        return min(trait_score, 100)
```

Method 2: Minimum Threshold + Average

```python
def calculate_trait_score_threshold(persona_scores, combination, min_threshold=30):
    """
    All personas in combination must be >30, then average
    """
    combo_scores = [persona_scores[combo] for combo in combination]

    if all(score >= min_threshold for score in combo_scores):
        return sum(combo_scores) / len(combo_scores)
    else:
        return 0  # Doesn't qualify for this trait
```

Trait Analysis Output Table:

UIN	full_name	trait_combination	trait_score	constituent_scores	trait_strength
D001	Dr. John Smith	Academic + Recognition	81.4	Academic:85, Recognition:72	Strong
D001	Dr. John Smith	Academic + Patient	79.1	Academic:85, Patient:67	Strong
D002	Dr. Priya Sharma	Patient + Community	85.1	Patient:92, Community:73	Very Strong

Trait Strength Classification:

- **Very Strong**: Trait score >80
- **Strong**: Trait score 65-80
- **Moderate**: Trait score 45-65
- **Weak**: Trait score 30-45
- **Not Applicable**: Trait score <30

PnC Generation Logic:

```python
```

```
# Generate all meaningful combinations
meaningful_combinations = []

# 2-persona combinations (C(7,2) = 21 combinations)
for i in range(len(personas)):
    for j in range(i+1, len(personas)):
        meaningful_combinations.append([personas[i], personas[j]])

# 3-persona combinations (only if all >40 points)
for i in range(len(personas)):
    for j in range(i+1, len(personas)):
        for k in range(j+1, len(personas)):
            combo = [personas[i], personas[j], personas[k]]
            if all(persona_scores[p] > 40 for p in combo):
                meaningful_combinations.append(combo)
```

4. Export Functionality

Export Options:

Option 1: Basic Persona Scores Export

Filename: `doctor_persona_scores_YYYY-MM-DD.csv`

```csv
UIN,full_name,specialty,dr_city,dr_state,academic_score,incentive_score,patient_score,technology_score,recognition_sco
```

Option 2: Filtered Results Export

Filename: `filtered_doctors_[filter_description]_YYYY-MM-DD.xlsx`

- Sheet 1: Filtered Doctor List
- Sheet 2: Filter Summary & Statistics
- Sheet 3: Persona Distribution Charts (data for charts)

Option 3: Trait Analysis Export

Filename: `trait_analysis_YYYY-MM-DD.xlsx`

- Sheet 1: All Trait Combinations
- Sheet 2: Strong Traits Only (score >65)
- Sheet 3: UIN-wise Trait Summary
- Sheet 4: Trait Statistics

Option 4: Complete Profile Export (for selected UIns)

Filename: `detailed_profiles_YYYY-MM-DD.xlsx`

- Sheet 1: Persona Scores
- Sheet 2: Professional Details
- Sheet 3: Research & Academic Data
- Sheet 4: Digital Presence Data
- Sheet 5: Contact Information

Export Features:

```
Export Filters:
├── Data Range: Current Filter / All Data / Custom Selection
├── Fields: All Fields / Persona Scores Only / Custom Fields
├── Format: CSV / Excel (.xlsx) / JSON
└── Include: Raw Scores / Percentile Ranks / Both

Advanced Options:
├── Include Charts & Visualizations (Excel only)
├── Add Data Dictionary Sheet
├── Include Filter Summary
└── Password Protection (optional)
```

5. User Interface Mockup

Main Dashboard Layout:

```
┌─────────────────────────────────────────────────────────────┐
│ Healthcare Professional Persona Analytics - Phase 1          │
├─────────────────────────────────────────────────────────────┤
│ FILTERS:                                                     │
│ [State ▼] [City ▼] [Specialty ▼] [Experience ▼]              │
│ [Primary Persona ▼] [Confidence ▼] [Reset Filters]           │
│                                   [Export ▼]                 │
├─────────────────────────────────────────────────────────────┤
│ RESULTS: 1,247 doctors found | Sort by: [Primary Persona ▼]  │
├─────────────────────────────────────────────────────────────┤
│ UIN   │ Name          │ Specialty │ Academic │ Patient │...│
│ D0001 │ Dr. John Smith │ Cardiology │   85   │   67   │...│
│ D0002 │ Dr. Priya Sharma │ Pediatrics │  34   │   92   │...│
│ D0003 │ Dr. Raj Kumar  │ Orthopedic │   72   │   45   │...│
│       │               │           │          │         │   │
└─────────────────────────────────────────────────────────────┘
```

```
┌─────────────────────────────────────────────────────────────┐
│ TRAIT ANALYSIS (PnC Combinations)                            │
├─────────────────────────────────────────────────────────────┤
│ Select Combination: [Academic ☑] [Patient ☑] [Technology ☐]  │
│ Minimum Trait Score: [65________] Calculate                  │
├─────────────────────────────────────────────────────────────┤
│ UIN   │ Name          │ Trait Score │ Strength  │ View       │
│ D0001 │ Dr. John Smith │   81.4     │ Strong    │ [Details] │
│ D0015 │ Dr. Lisa Wong  │   76.8     │ Strong    │ [Details] │
└─────────────────────────────────────────────────────────────┘
```

6. Implementation Priority

Week 1-2: Core Scoring System

- UIN-centric data structure
- Basic persona scoring algorithms
- Primary/secondary persona assignment

Week 3: Filter & Search System

- Geographic and professional filters
- Score range filters
- Confidence level filtering

Week 4: Trait Analysis (PnC)

- Combination generation logic
- Trait scoring algorithms
- Trait strength classification

Week 5: Export Functionality

- CSV/Excel export options
- Filtered export capability
- Data formatting and validation

Week 6: Testing & Refinement

- User acceptance testing
- Performance optimization
- Documentation and training materials

Success Metrics for Phase 1:

- ✅ All doctors have persona scores calculated
- ✅ Filter system returns results in <2 seconds
- ✅ PnC combinations identify meaningful doctor segments
- ✅ Export functionality works for datasets up to 10,000 doctors
- ✅ User can find specific doctor profiles in <30 seconds